

Robin Schäfer

Dr. rer. nat.

✉ robin_schaefer@fas.harvard.edu
🌐 robin-schaefer.github.io
ID 0000-0001-9728-2371
🔗 NX7j0dsAAAAJ

Dissertation

R. Schäfer

Magnetic frustration in three dimensions

Dissertation published via TU Dresden (2022)

Publication list

- [17] **R. Schäfer**, C. Chamon, C. R. Laumann
Hall-on-Toric: Descendant Laughlin state in the chiral Z_p toric code
[Phys. Rev. Lett. 136, 166603 \(2026\)](#)
- [16] **R. Schäfer**, P. Ebert, N. Hassan, J. Reuther, D. Luitz, A. Wietek
Thermodynamics of the Heisenberg antiferromagnet on the maple-leaf lattice
[Zeitschrift für Naturforschung A. \(zna-2025-0382\)](#)
- [15] D. Vuina, **R. Schäfer**, D. M. Long, A. Chandran
Probing Hilbert space fragmentation using controlled dephasing
[Phys. Rev. B 112, 134305 \(2025\)](#)
- [14] E. M. Smith, A. Fitterman, **R. Schäfer**, et al.
Two-Peak Heat Capacity Accounts for $R\ln(2)$ Entropy and Ground State Access in the Dipole-Octupole Pyrochlore $Ce_2Hf_2O_7$
[Phys. Rev. Lett. 135, 086702 \(2025\)](#) — Editor's Suggestion
- [13] E. M. Smith, **R. Schäfer**, et al.
Single Crystal Diffuse Neutron Scattering Study of the Dipole-Octupole Quantum Spin Ice Candidate $Ce_2Zr_2O_7$: No Apparent Octupolar Correlations Above $T=0.05K$
[Phys. Rev. X 15, 021033 \(2025\)](#)
- [12] **R. Schäfer**, and D. J. Luitz
DanceQ: High-performance library for number conserving bases
[SciPost Phys. Codebases 48 \(2025\)](#)
[DanceQ repository](#) and [DanceQ documentation](#)
- [11] Z. Lu, **R. Schäfer**, J. N. Hallén, C. R. Laumann
[111]-strained spin ice: Localization of thermodynamically deconfined monopoles
[Phys. Rev. B 110, 184421 \(2024\)](#)

- [10] D. Yahne, B. Placke, **R. Schäfer**, *et al.*
Dipolar spin ice regime proximate to an all-in-all-out Néel ground state in the dipolar-octupolar pyrochlore $\text{Ce}_2\text{Sn}_2\text{O}_7$
[Phys. Rev. X 14, 011005 \(2024\)](#)

- [9] J. Beare, E. M. Smith, J. Dudemaine, **R. Schäfer**, *et al.*
 μSR Study of the Dipole-Octupole Quantum Spin Ice Candidate $\text{Ce}_2\text{Zr}_2\text{O}_7$
[Phys. Rev. B 108, 174411 \(2023\)](#)

- [8] E. M. Smith, J. Dudemaine, B. Placke, **R. Schäfer**, *et al.*
Quantum Spin Ice Response to a Magnetic Field in the Dipole-Octupole Pyrochlore $\text{Ce}_2\text{Zr}_2\text{O}_7$
[Phys. Rev. B 108, 054438 \(2023\)](#)

- [7] **R. Schäfer**, B. Placke, O. Benton, and R. Moessner
Abundance of hard-hexagon crystals in the quantum pyrochlore antiferromagnet
[Phys. Rev. Lett. 131, 096702 \(2023\)](#)

- [6] **R. Schäfer**, J. C. Budich, and D. J. Luitz
Symmetry protected exceptional points of interacting fermions
[Phys. Rev. Research 4, 033181 \(2022\)](#)

- [5] I. Hagymási, **R. Schäfer**, R. Moessner, and D. J. Luitz
Magnetization process and ordering of the $S = 1/2$ pyrochlore Heisenberg antiferromagnet in a magnetic field
[Phys. Rev. B 106, L060411 \(2022\)](#)

- [4] E. Smith, O. Benton, D. Yahne, B. Placke, **R. Schäfer**, *et al.*
The case for a $U(1)\pi$ Quantum Spin Liquid Ground State in the Dipole-Octupole Pyrochlore $\text{Ce}_2\text{Zr}_2\text{O}_7$
[Phys. Rev. X 12, 021015 \(2022\)](#)
 — **Featured in** [Quantum Science and Technology Collection](#)

- [3] I. Hagymási, **R. Schäfer**, R. Moessner, and D. J. Luitz
Possible Inversion Symmetry Breaking in the $S = 1/2$ Pyrochlore Heisenberg Magnet
[Phys. Rev. Lett. 126, 117204 \(2021\)](#)

- [2] **R. Schäfer**, I. Hagymási, R. Moessner, and D. J. Luitz
Pyrochlore $S = \frac{1}{2}$ Heisenberg antiferromagnet at finite temperature
[Phys. Rev. B 102, 054408 \(2020\)](#)

- [1] **R. Schäfer**, G. S. Uhrig, and J. Stolze
Time-crystalline behavior in an engineered spin chain
[Phys. Rev. B 100, 184301 \(2019\)](#)

Preprints

- [6] A. Sanders, G. Naik, J. Hallén, **R. Schäfer**
Defect Poisoning of Quantum Spin Ice
[arXiv:2609.28643 \(2026\)](#)
- [5] J. S. Herz, **R. Schäfer**, M. G. Gonzalez, D. J. Luitz
Sign-optimized Quantum Monte Carlo
[arXiv:2607.24679 \(2026\)](#)
- [4] S. Raman, **R. Schäfer**, A. J. Kollar, A. Chandran
— **shared first authorship**
Dissipative Stabilization of Floquet-Engineered Many-Body Order
[arXiv:2607.16391 \(2026\)](#)
- [3] E. M. Smith, M. S. Powell, **R. Schäfer**, et al.
Zero-Field Long Range Order at $T \sim 40$ mK in the Proximate Quantum Spin Ice $\text{Ce}_2\text{Sn}_2\text{O}_7$ and Phase Diagram for Magnetic Fields Along $[1,1,0]$
[arXiv:2607.12274 \(2026\)](#)
- [2] Bo Yuan, M. Powell, X. Liu, J. Ni, E. M. Smith, **R. Schäfer**, et al.
Observation of Dipolar Spin-ice-like Correlations in the Quantum Spin Ice Candidate $\text{Ce}_2\text{Sn}_2\text{O}_7$
[arXiv:2601.20766 \(2026\)](#)
- [1] H. Kim, **R. Schäfer**, D. M. Long, A. Polkovnikov, A. Chandran
— **shared first authorship**
Confined and deconfined chaos in classical spin systems
[arXiv:2507.07168 \(2025\)](#)